

Politecnico di Torino
Dipartimento di Automatica e Informatica
Deep NLP (01VIXSM)
Written Exam simulation

January 16th, 2024

Name: _____

Surname: _____

Student ID: _____

Exam rules:

- The present exam consists of 6 pages (including this cover page) and 7 questions overall. Any inconsistencies/printing errors in the written exam content must be reported to the teacher *at the beginning* of the exam.
- Exam duration: *60 minutes*.
- Withdraw is allowed only *at the end* of the exam.
- The exam is *closed-book*. Electronic devices, mobile phones, smart watches, and extra papers (even blank papers) are *not allowed*.
- Closed-ended questions: cross the right answer (just one) at pag. 2. Wrong or missing answers to closed-ended questions will receive *no penalty*.
- Open questions: write your answers below the text of the question. If you need more space please use the last page (i.e., pag. 6) and/or the back side of the paper.

Evaluation grid

Question:	1	2	3	4	5	6	7	Total
Points:	1	1	1	1	4	6	8	22
Score:								

1. (1 point) What of the following statement holds **True** for the Word2Vec model?
 - A model with larger context window takes longer time to train.
 - Stopword removal does not influence the model behaviour.
 - The standard training procedures rely on annotated data.
 - It is a Deep Learning model.
 - The model size is independent on the corpus used for training.
2. (1 point) A deep learning-based sentence embedding model
 - Has the goal of generating a vector for each word composing the sentence.
 - Can be used for tackling Information Retrieval tasks.
 - Generates a syntactic-aware representation without considering the semantics of the text.
 - Does not require training.
 - None of the above.
3. (1 point) Given a query Q , the recall of a Information Retrieval system
 - Measures the fraction of retrieved outcomes that are relevant to the requested requested information.
 - Is the ratio of the number of non-relevant documents that are retrieved to all non-relevant documents available.
 - Measures the fraction of the relevant documents in the collection that are returned by the system.
 - Measures the average of the reciprocal ranks of results for Q .
 - None of the above.
4. (1 point) A Named Entity Recognition system:
 - Is able to translate named entities.
 - Can be used to identify location references.
 - Capitalizes entities according to the user needs.
 - Annotates Noun, Verb, Pronoun and adjective from the given sentence tokens.
 - Summarizes the topics related to a give entity.

5. (4 points) Describe the concept of **stopword**:

1. Define the concept.
2. Provide some examples in the English language.
3. Report and explain at least two examples of proper handling of stopwords in NLP systems.

5.1: Stopwords are a set of commonly used words in a given language. They are commonly used in NLP to remove common words that do not provide any useful informative content.

5.2: Examples in English are **an, of, is**.

5.3:

Example 1: Statistics-based IR systems need to focus on the words relevant to the given query. Thus, they commonly remove stopwords from the query text.

E.g., The cat sat on the mat - *cat sat mat*.

Notice that transformer-based IR systems do not require stopwords elimination.

Example 2: Statistics-based sentiment analysis methods remove stopwords to avoid introducing noise or bias. In this specific case, *not* must not be removed to avoid misclassifications due to negations.

E.g., This new phone is so good - *new phone good*

E.g., This new phone is not so good - *new phone not good*

6. (6 points) Considering the Word2Vec model

1. Describe its inner working and main goals.
2. Illustrate in details one of the training procedures (at your choosing).
3. Discuss (at least) one of the main drawbacks of the model.
4. Enumerate the key differences with FastText.

6.1:

- The word embedding model generates one vector representation per word in the vocabulary.
- Word vectors are not contextualized (unlike, e.g., ELMO)
- It relies on self-supervised training using sliding window.
- Architectures: Skip-gram and CBOW.
- The output of the model is a matrix containing the mapping between each word in the vocabulary and its corresponding vector.

6.2: The CBOW training procedure:

- The context is represented as a bag of the words contained in a fixed-size window aligned with the target word.
- The distributed representations of the context is used to predict the word in the middle of the window.
- The classification model exploits the context to predict the correct target word.
- *Sketch of the architecture.*

6.3:

Limitation: word representations are not contextual.

The cat eat the **mouse**¹

The **mouse**² is not working anymore.

$$W2V(\text{mouse}^1) = W2V(\text{mouse}^2)$$

6.4:

FastText creates n-gram embeddings and generates word vectors using compositionality.

E.g. (with n-gram = 4),

$$V(\text{processing}) = FT(<\text{pro}) + FT(\text{proc}) + FT(\text{roce}) + \dots + FT(\text{ing}) + FT(\text{processing})$$

7. (8 points) The *Polito* company need to identify critical reviews crawled from the web. Reviews are not necessarily written in English. Let us consider a list of text snippets containing the reviews as input. Address the following points:

1. Define a NLP pipeline leveraging on pre-trained models. Let us suppose that no annotations are provided at this stage.
2. In view of the positive feedbacks received from your company supervisor, you are now allowed to ask for 5K annotations. Describe the annotations you are willing to collect and how they can be used to improve the performance of your model.

7.1 Main pipeline steps (example):

- Language identification.
- Pretrained machine translation model applied to non-target language reviews.
- Train sentence embedding model on different sentiment-aware datasets or use a pre-trained model to map reviews into vector embeddings.
- Apply clustering to distinguish the 2 clusters.
- Manually inspect the clusters decide which one contains the critical reviews.

7.2

1. Sampling from different clusters choose 5K data sentences.
2. Let human annotator label the reviews in the 5K sampled data as critical or non-critical
3. Train a supervised classification model to perform binary classification.
4. Use Train/Test/Validation (80%/10%/10%) split. Validation set is used for stopping model training once overfitting is found.
5. Assess the model performance.

Optional (prior to Step 2): Get an higher amount of samples from near cluster borders.

This page is intentionally left blank to accommodate work that wouldn't fit elsewhere and/or scratch work.